



Sound

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What is sound?

Key takeaway:

- Sound takes up space

```
Loading required package: sysfonts
```

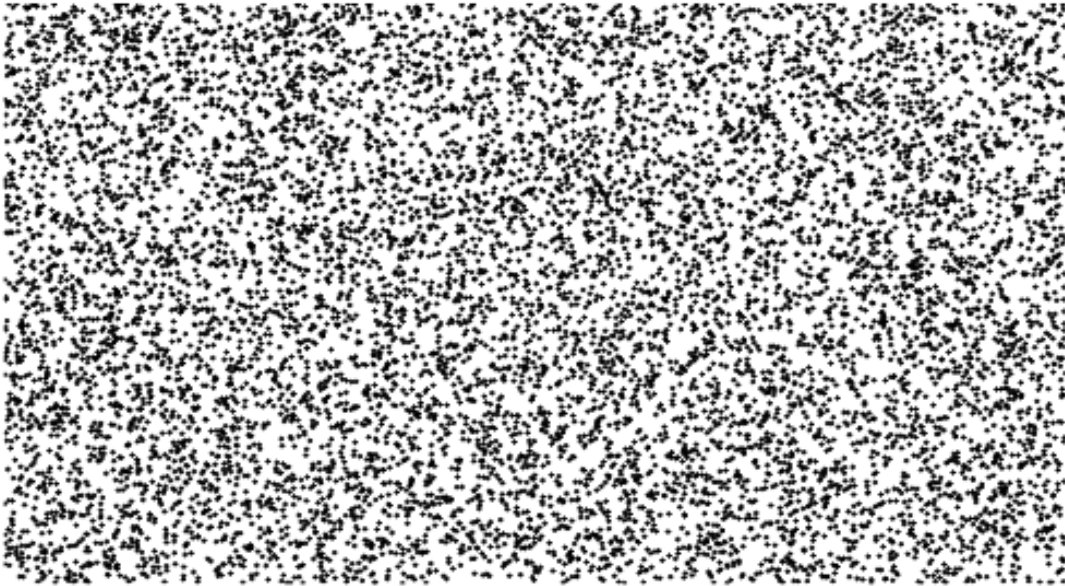
```
Loading required package: showtextdb
```

```
— Attaching core tidyverse packages — tidyverse 2.0.0 —
✓ dplyr      1.2.1      ✓ readr      2.2.0
✓ forcats    1.0.1      ✓ stringr    1.6.0
✓ lubridate  1.9.5      ✓ tibble     3.3.1
✓ purrr      1.2.2      ✓ tidyr      1.3.2
— Conflicts — tidyverse_conflicts()
—
✗ dplyr::filter() masks stats::filter()
✗ dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
conflicts to become errors
```



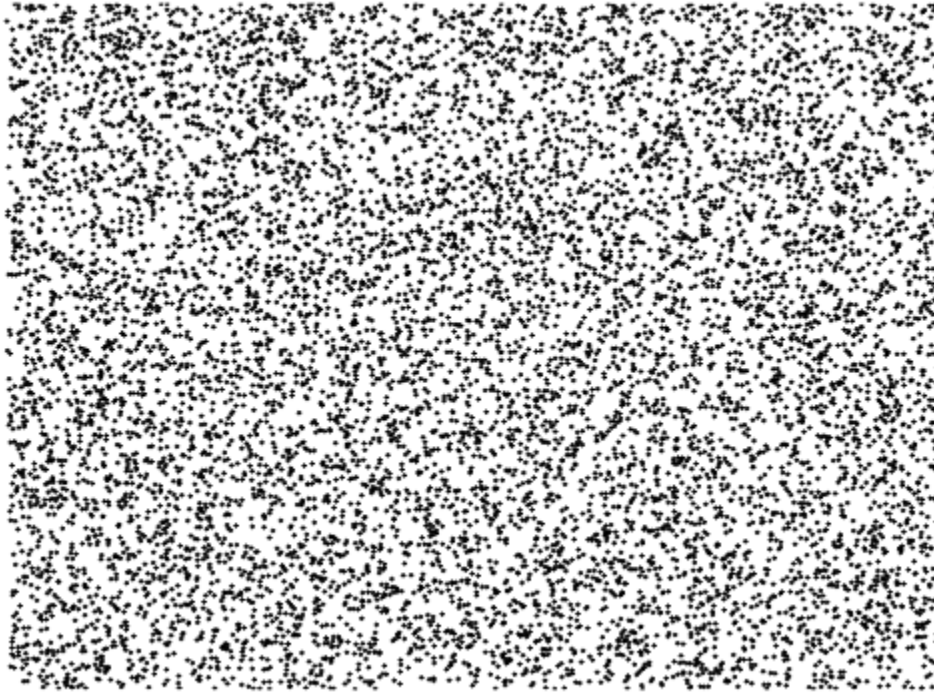
Not Sound

air



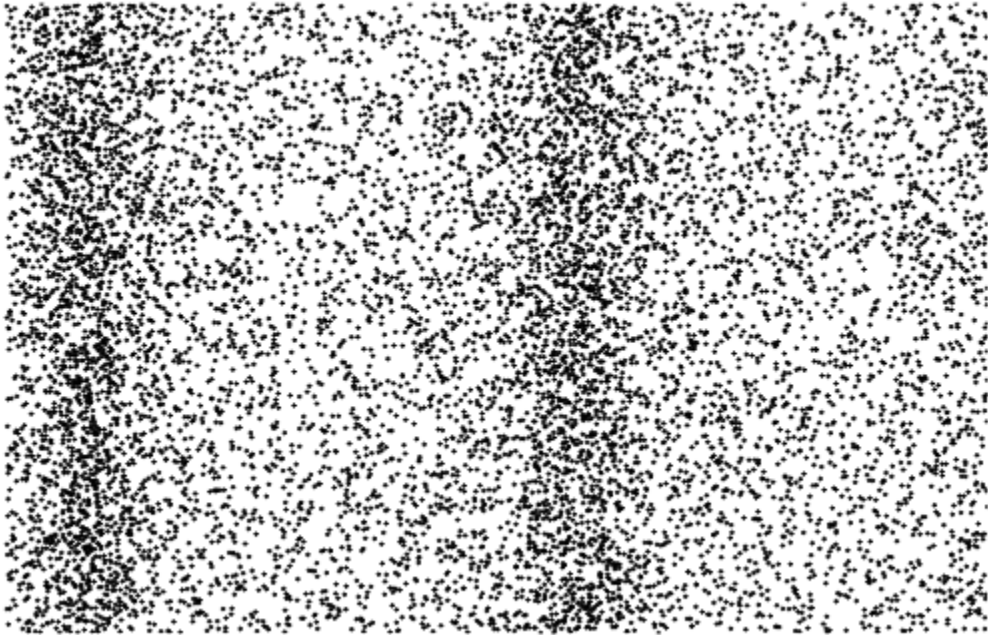


Sound - Aperiodic

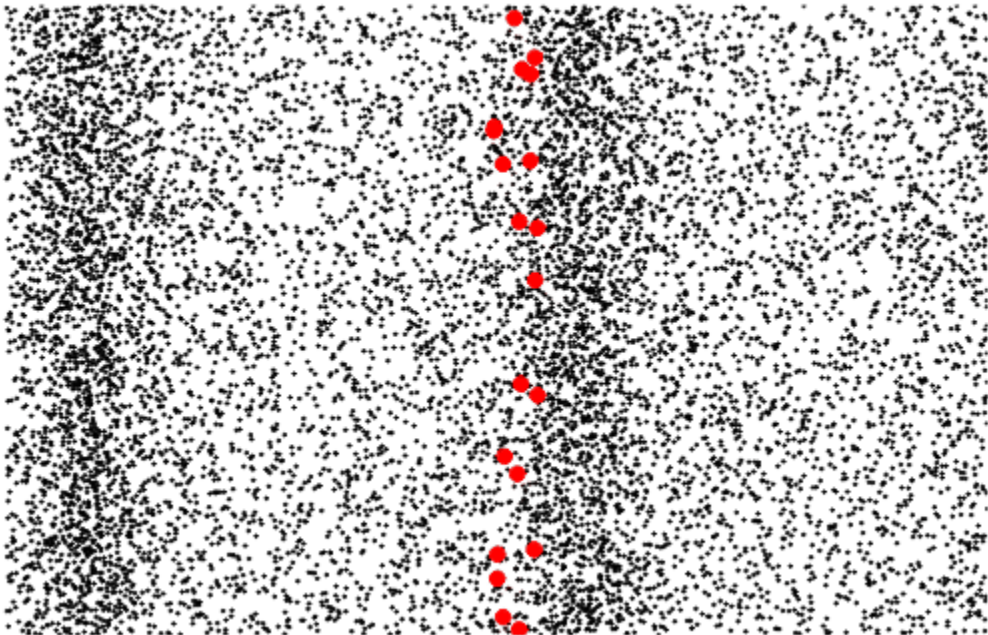




Sound - Periodic



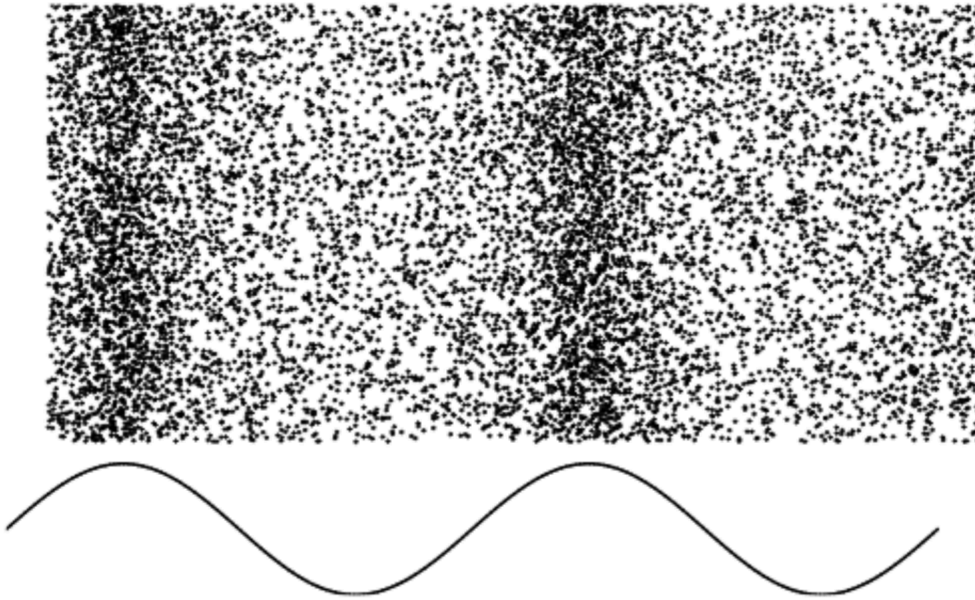
Sound





Sound is not the air molecules from my lungs hitting your ears.

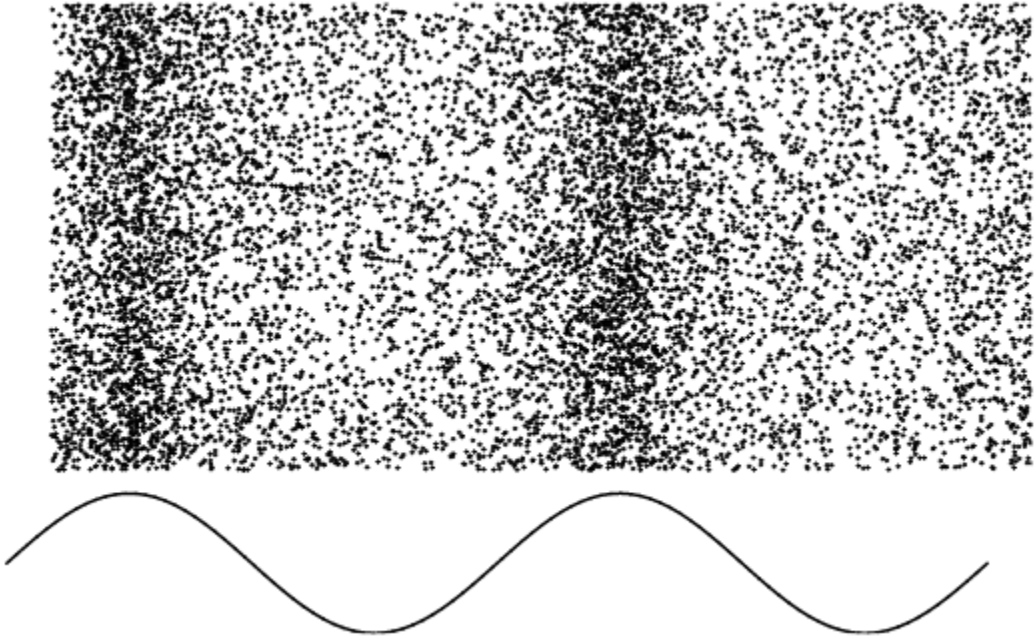
Sound



Where air is tightly packed, there's high pressure. Where it's more spread out, low pressure.

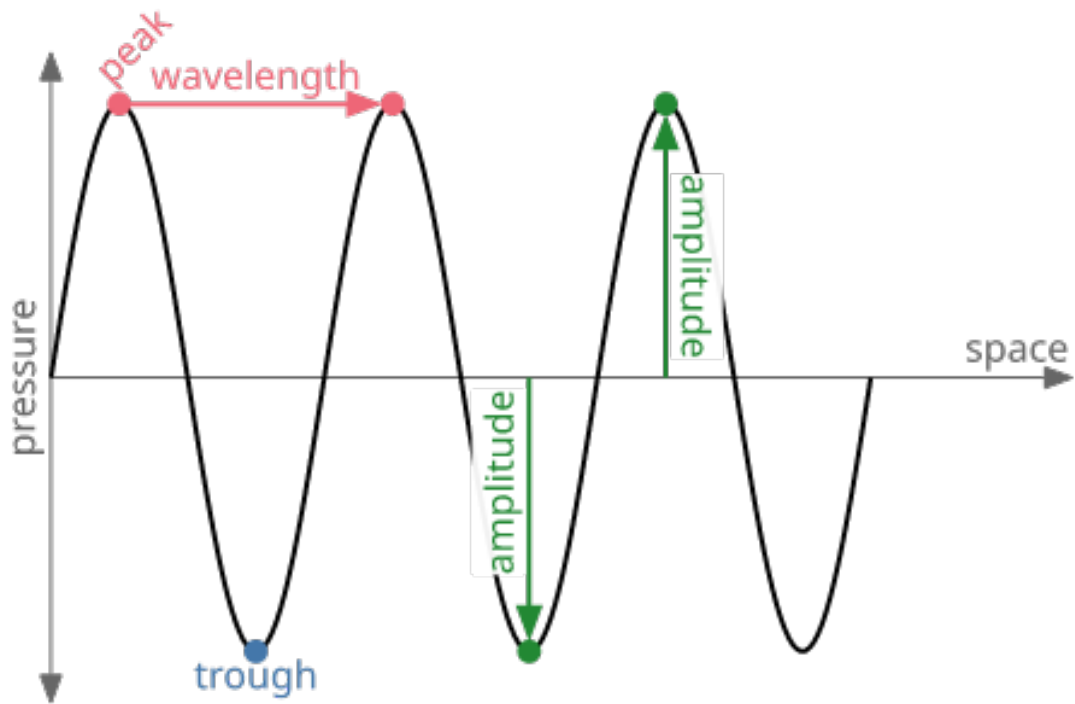


Sound

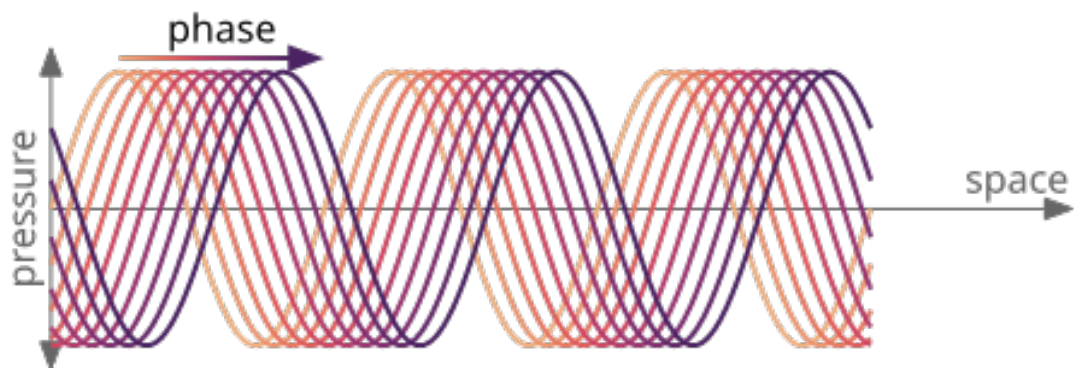




Sound

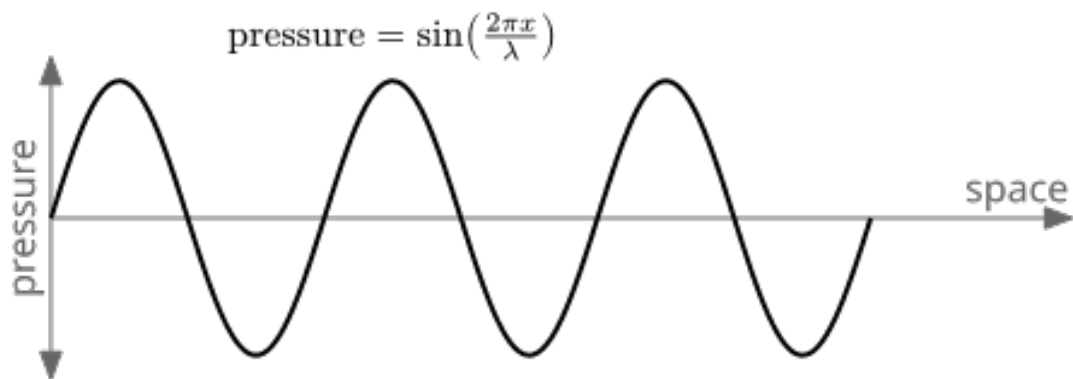


Sound





Sound - Sine Wave



Sounds - Frequency

Frequency = how many high pressure zones per second

This is going to be a function of both the wavelength and the speed of the wave.

Because the speed of sound will usually be fixed, Wavelength $\propto$ Frequency.

Frequency to Wavelength

$$\text{speed of sound} \approx 34300 \frac{\text{cm}}{\text{s}}$$

$$f \text{ Hz} = \frac{f \text{ cycle}}{\text{s}}$$

Frequency to Wavelength

$$\begin{aligned} \text{wavelength} &= \frac{34300 \text{ cm}}{\text{s}} \frac{1}{f \text{ Hz}} \\ &= \frac{34300 \text{ cm}}{\text{s}} \frac{\text{s}}{f \text{ cycle}} \\ &= \frac{34300 \text{ cm}}{f \text{ cycle}} \end{aligned}$$

Frequency to Wavelength

$$\begin{aligned} &34300 \frac{\text{cm}}{\text{s}} \\ 440 \text{ Hz} &= \frac{440}{\text{s}} \end{aligned}$$

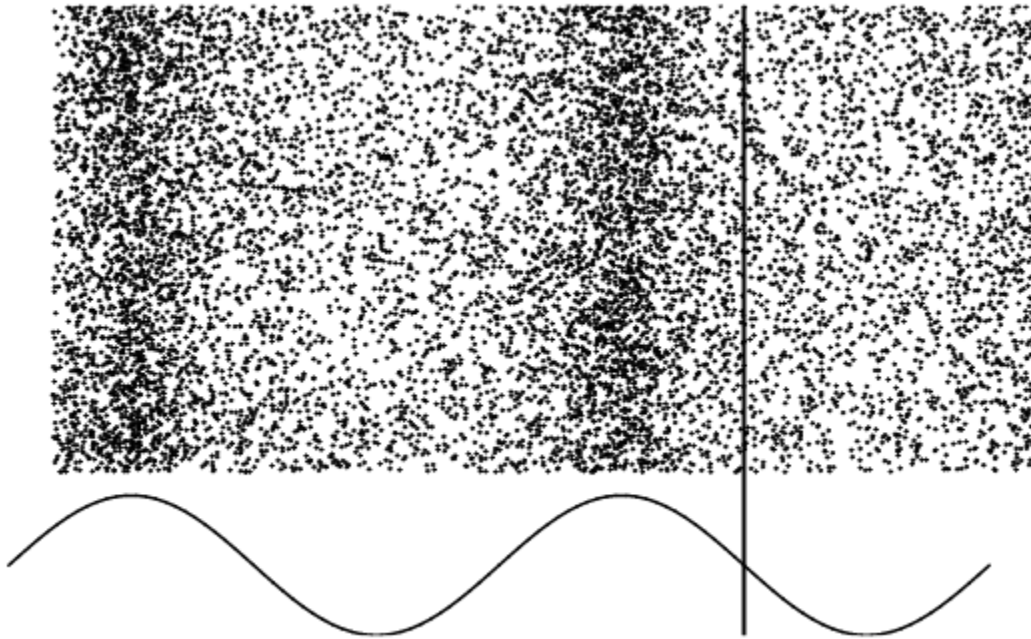


$$\frac{34300 \text{ cm}}{s} \frac{s}{440} \approx 78 \text{ cm} \approx 2.6 \text{ ft}$$

Wavelength to Frequency

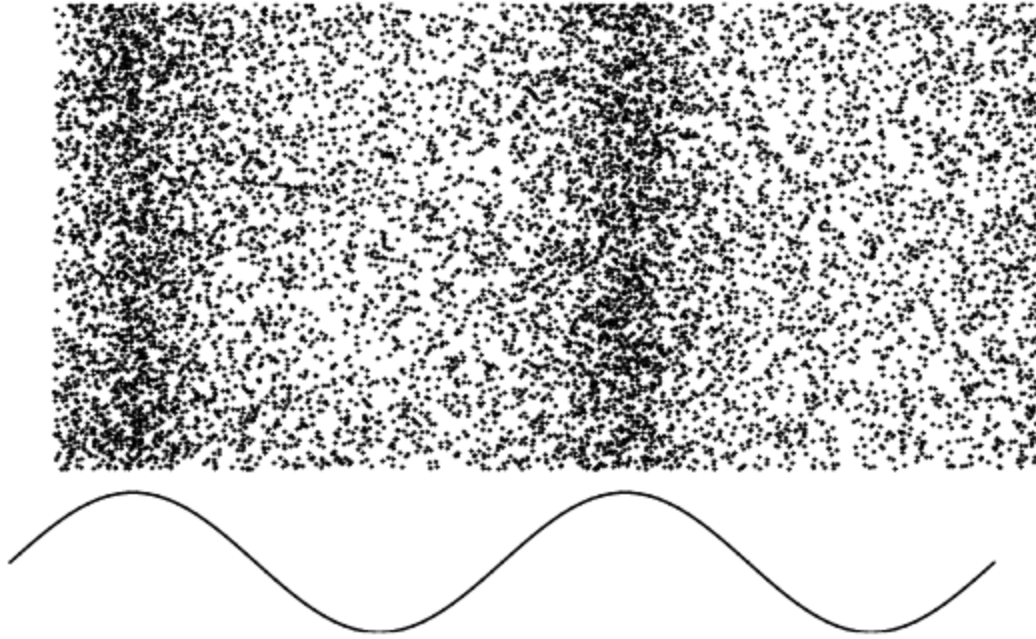
Sounds - Frequency

```
Joining with `by = join_by(t)`
```



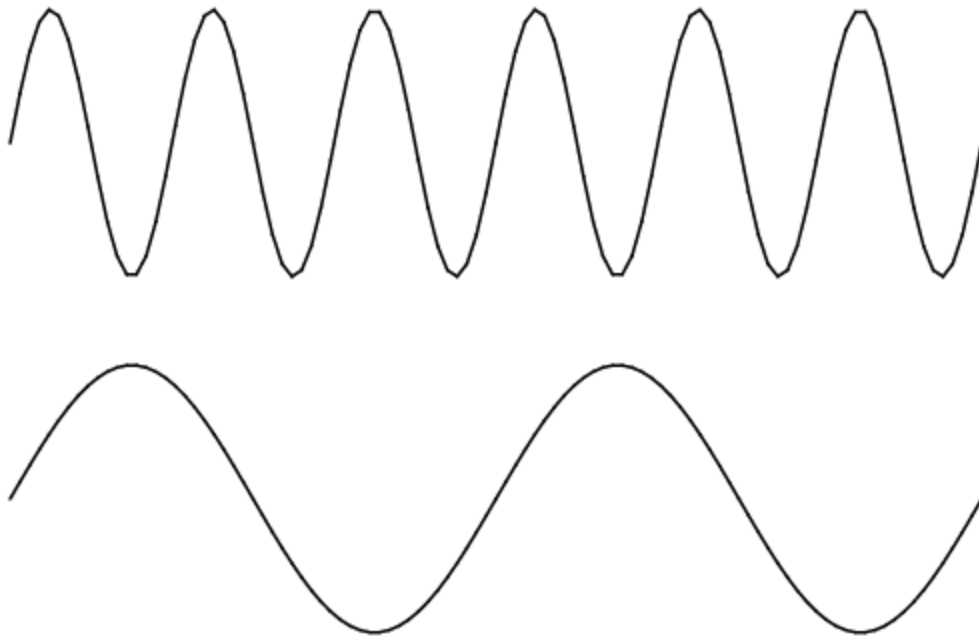
Sound - Increasing Frequency

As wavelength decreases, frequency increases



Sound - Frequency

The *speed* of these higher frequency waves is the *same* as the lower frequency waves

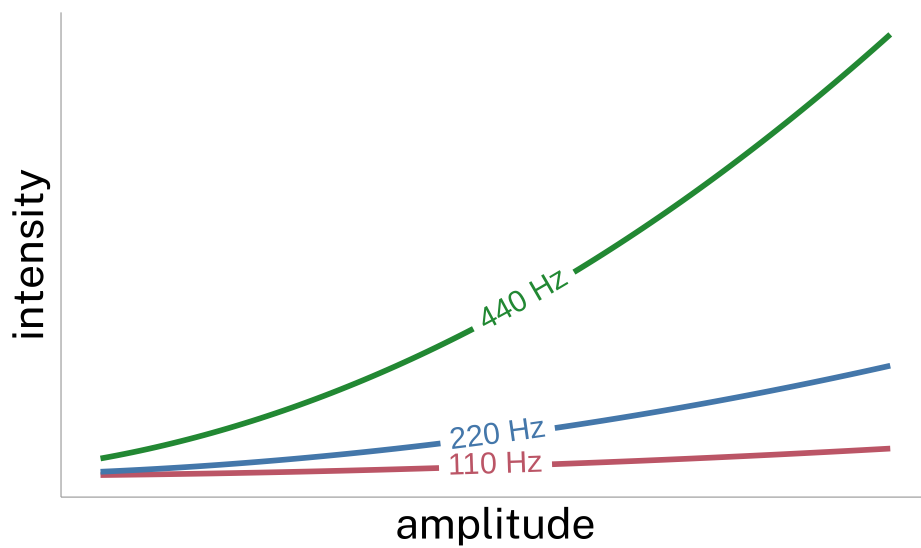


Sound - Frequency

Every time frequency doubles, we experience it as the frequency increasing 1 octave.

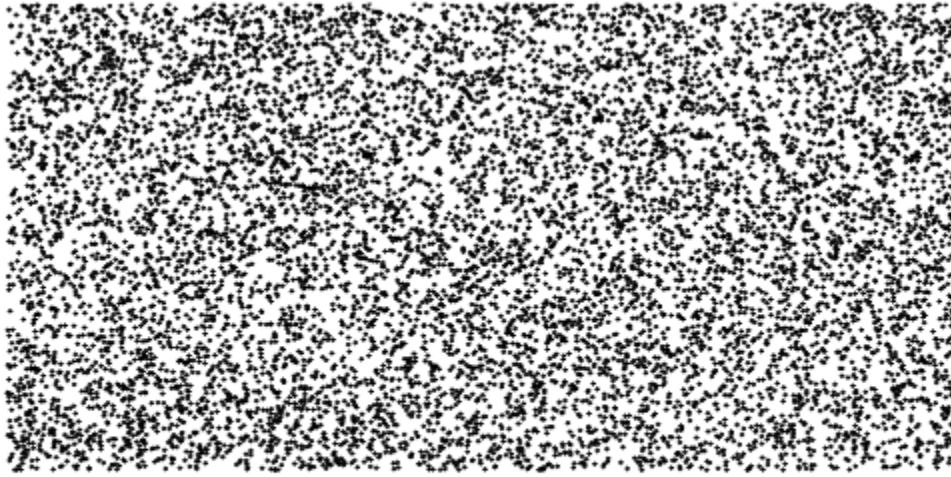
Sound - Intensity

A relationship between frequency *and* amplitude





Sound - Increasing Amplitude



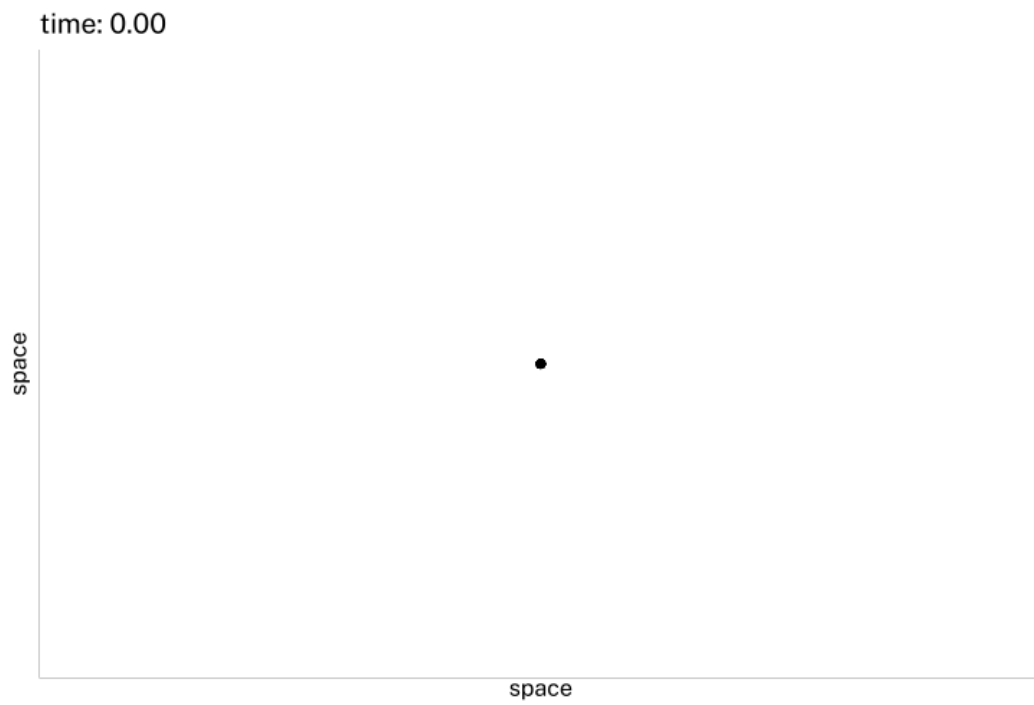
Sound - Loudness

Decibels (dB) are a measure of *relative* intensity.

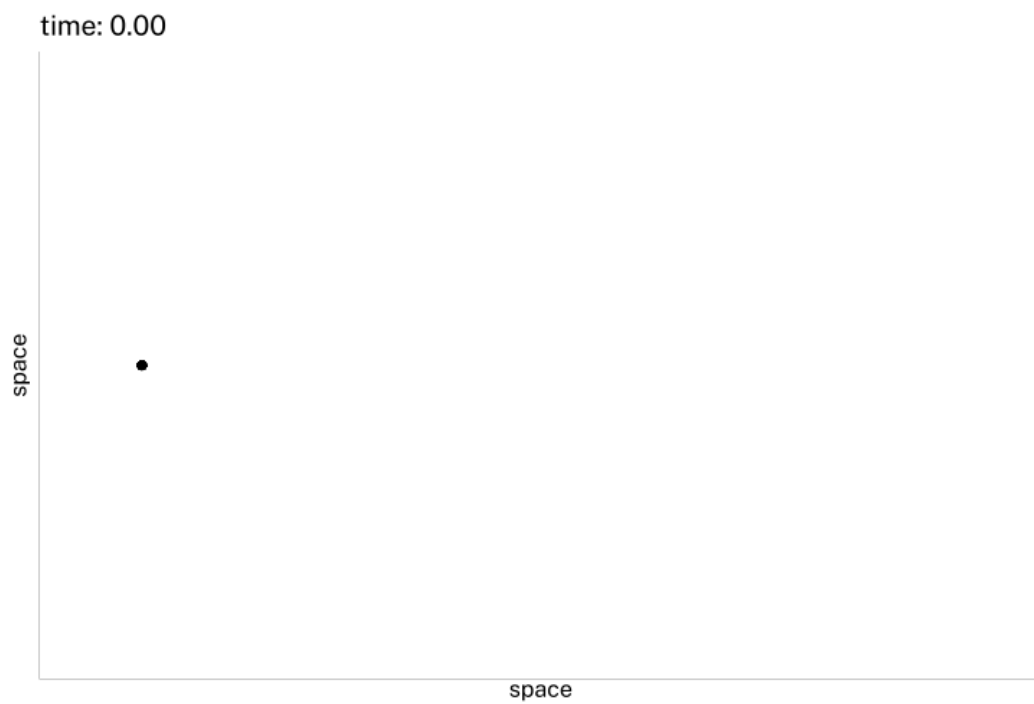
$$1dB = 10 \log_{10} \left(\frac{s_1}{s_2} \right)$$



Sound - The Doppler Effect



Sound - The Doppler Effect





The Doppler Effect

```
Warning: Removed 15 rows containing missing values or values outside the scale range  
(`geom_line()`).
```

